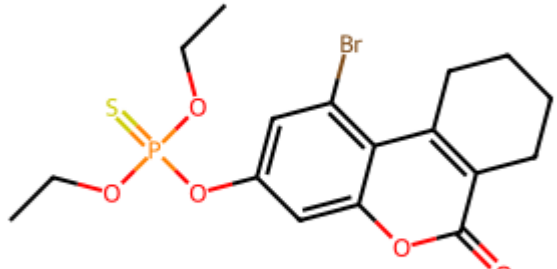


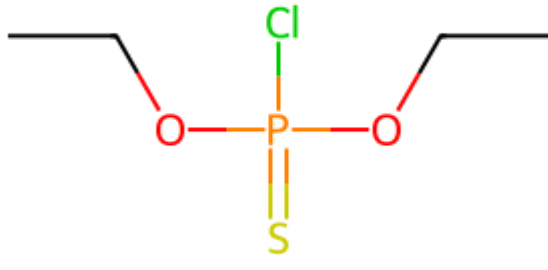
{'analysis_time': 0.0,
'first_solution_time': 0.10049,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.01005033585350145,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.01005033585350145,
'retro_star_value': 0.01005033585350145}

Reaction
{'analysis_time': 0.10049,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 26.11894950399772,
'retro_star_reaction_number': 22.87872670354909,
'retro_star_rxn_cost': 11.512925464970229,
'retro_star_value': 22.87872670354909}

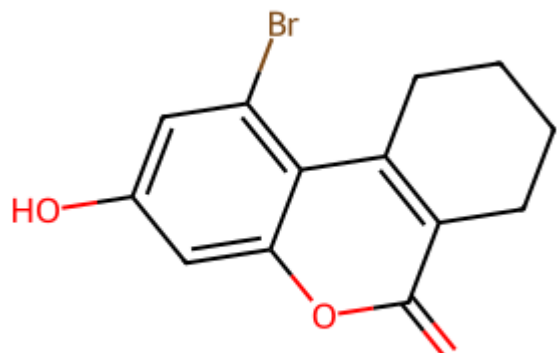


{'analysis_time': 0.10049,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 14.606024039027492,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 11.365801238578863,
'retro_star_value': 22.87872670354909}

Reaction
{'analysis_time': 2.081637,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 14,
'num_calls_value_function': 474,
'retro_star_min_cost': 14.606024039027492,
'retro_star_reaction_number': 11.365801238578863,
'retro_star_rxn_cost': 0.01005033585350145,
'retro_star_value': 22.87872670354909}

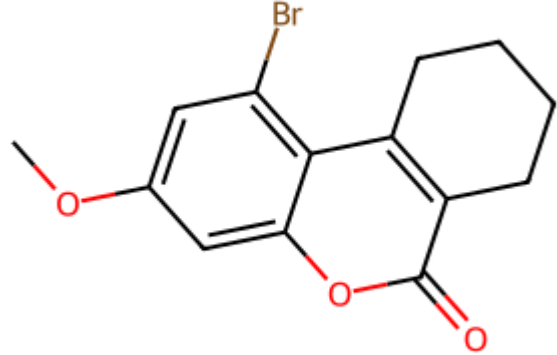


{'analysis_time': 2.081637,
'first_solution_time': 2.081637,
'num_calls_rxn_model': 14,
'num_calls_value_function': 474,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 22.87872670354909}



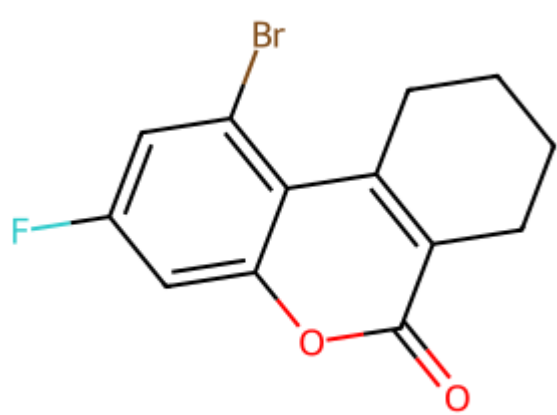
{'analysis_time': 2.081637,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 14,
'num_calls_value_function': 474,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 14.595973703173991,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 11.355750902725362,
'retro_star_value': 22.87872670354909}

Reaction
{'analysis_time': 3.346832,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 22,
'num_calls_value_function': 731,
'retro_star_min_cost': 14.595973703173991,
'retro_star_reaction_number': 12.431085695150994,
'retro_star_rxn_cost': 0.9181602301807653,
'retro_star_value': 23.954061495974724}



{'analysis_time': 3.346832,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 22,
'num_calls_value_function': 731,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 13.677813472993225,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 11.512925464970229,
'retro_star_value': 23.954061495974724}

Reaction
{'analysis_time': 5.593799,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 39,
'num_calls_value_function': 1253,
'retro_star_min_cost': 13.677813472993225,
'retro_star_reaction_number': 13.677813472993225,
'retro_star_rxn_cost': 7.008866593046887,
'retro_star_value': 26.118949503997722}



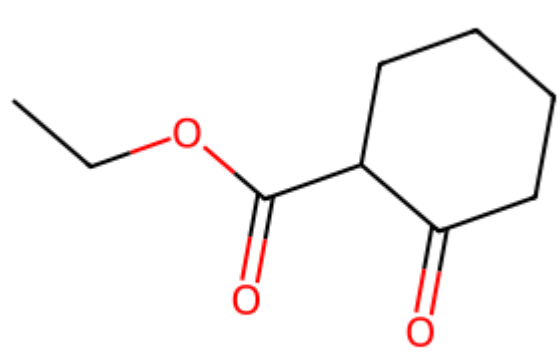
{'analysis_time': 5.593799,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 39,
'num_calls_value_function': 1253,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 6.455965904610345,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 6.455965904610345,
'retro_star_value': 26.118949503997722}



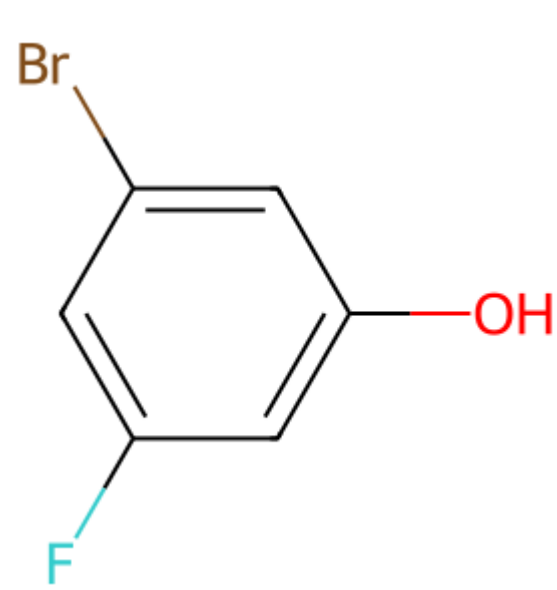
{'analysis_time': 5.593799,
'first_solution_time': 68.800219,
'num_calls_rxn_model': 39,
'num_calls_value_function': 1253,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.21298097533599225,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.21298097533599225,
'retro_star_value': 26.118949503997722}

Reaction
{'analysis_time': 73.590501,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 494,
'num_calls_value_function': 24413,
'retro_star_min_cost': 6.455965904610345,
'retro_star_reaction_number': 6.455965904610345,
'retro_star_rxn_cost': 6.455965904610345,
'retro_star_value': 26.118949503997722}

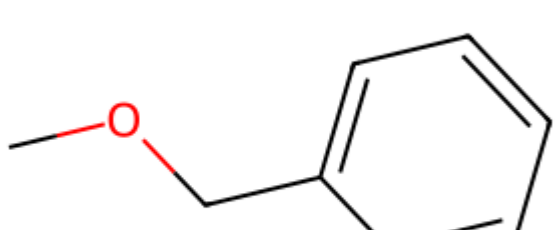
Reaction
{'analysis_time': 68.800219,
'first_solution_time': 68.800219,
'num_calls_rxn_model': 457,
'num_calls_value_function': 23055,
'retro_star_min_cost': 5.005921348433852,
'retro_star_reaction_number': 5.005921348433852,
'retro_star_rxn_cost': 5.005921348433852,
'retro_star_value': 30.911889877095582}



{'analysis_time': 73.590501,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 494,
'num_calls_value_function': 24413,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 26.118949503997722}



{'analysis_time': 73.590501,
'first_solution_time': 73.590501,
'num_calls_rxn_model': 494,
'num_calls_value_function': 24413,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 26.118949503997722}



{'analysis_time': 68.800219,
'first_solution_time': 68.800219,
'num_calls_rxn_model': 457,
'num_calls_value_function': 23055,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 30.911889877095582}